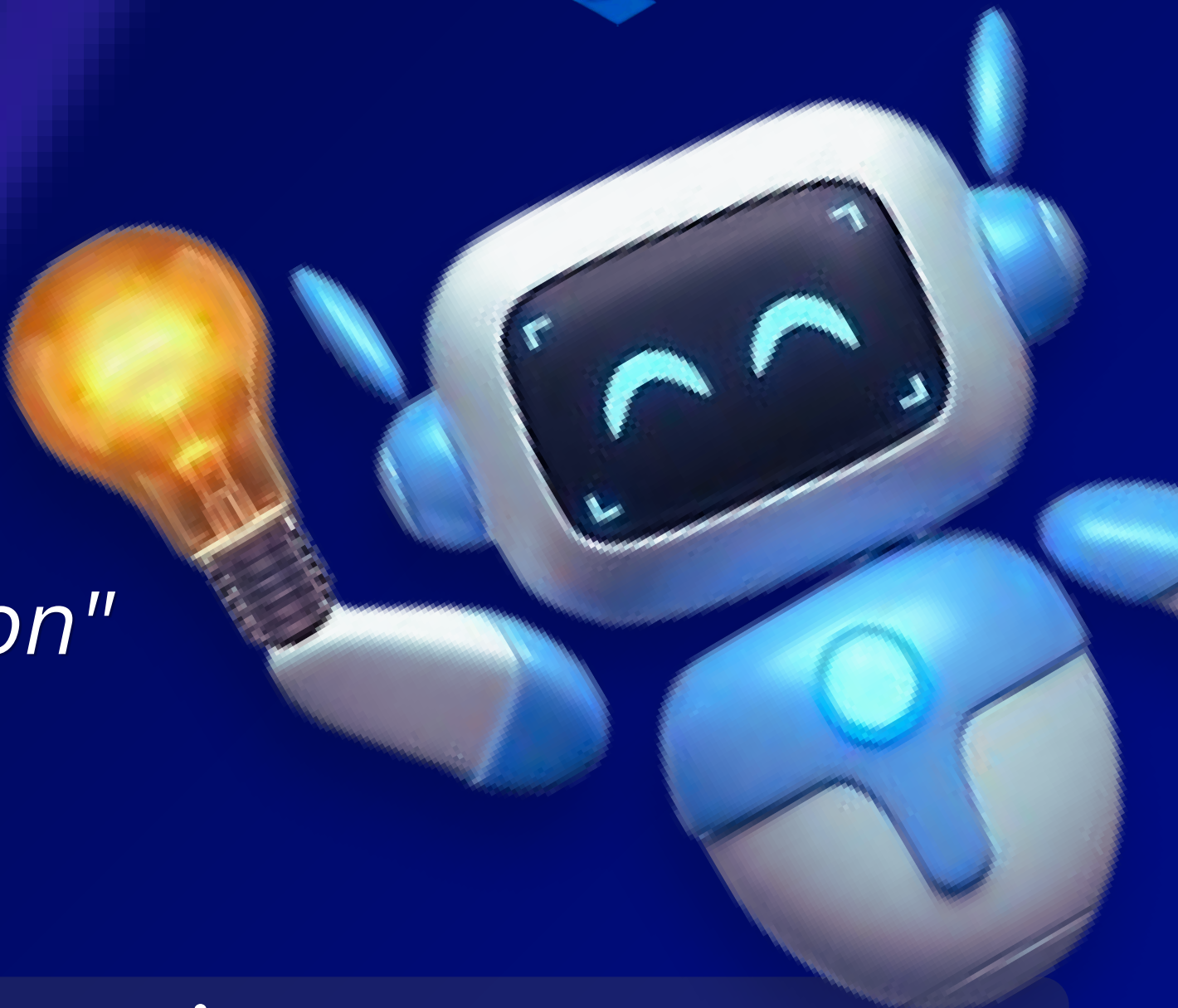


Smart Classroom Booking System (SCBS)

"Enhancing Campus Sustainability and Efficiency through IoT Integration"



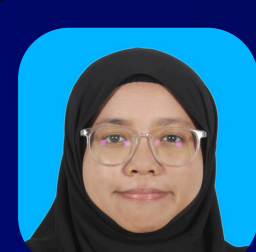
MEMBERS

- 

NORFAZLINA BINTI JOHAR
- 

NOR AZURA BINTI SALLEH
@ OMAR
- 

DR. MUHAMMAD BIN AMRAN
- 

NURUL AIN SHAZWANI BINTI
AHMAD ZAINI
- 

AISYAH NABILA BINTI MD YAZID
- 

PUTERI NUR ATHIRAH FARZANA
BINTI KAMARUZANAN

INTRODUCTION

The Smart Classroom Booking System (SCBS) is an innovative framework that modernizes campus management by integrating a Laravel-based web platform with ESP32 microcontrollers and PIR sensors to automate classroom scheduling, enhance security through QR access, and significantly reduce energy wastage.

KEY ISSUES

- 1

Manual Inefficiency: Paper-based logs cause double-bookings and delays.
- 2

Unauthorized Usage: Lack of access control leads to unscheduled room occupancy.
- 3

Energy Wastage: Utilities left active in empty rooms increase operational costs.
- 4

Information Gaps: Lack of live data makes finding available study space difficult.

OBJECTIVES

- 1

Digitize Scheduling: Replace manual logs with a centralized Laravel platform.
- 2

Secure Access: Implement QR code authentication for authorized entry only.
- 3

Optimize Energy: Use PIR sensors to automate utility control in empty rooms.
- 4

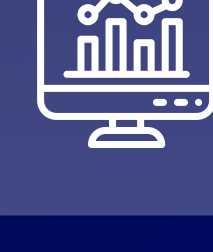
Provide Real-Time Visibility: Enable instant mobile/desktop tracking of room status.

KEY FEATURES

- 

Real-Time Dashboard: Instant "Available/Booked/In-Use" status tracking.
- 

QR Access Control: Secure, time-sensitive digital keys for entry.
- 

IoT Energy Automation: Auto-power off for lights/AC via motion detection.
- 

Admin Analytics: Automated usage reports and occupancy audit logs.

IMPACT & BENEFITS

- 

Eliminates Conflicts : Ends double-bookings and administrative delays.
- 

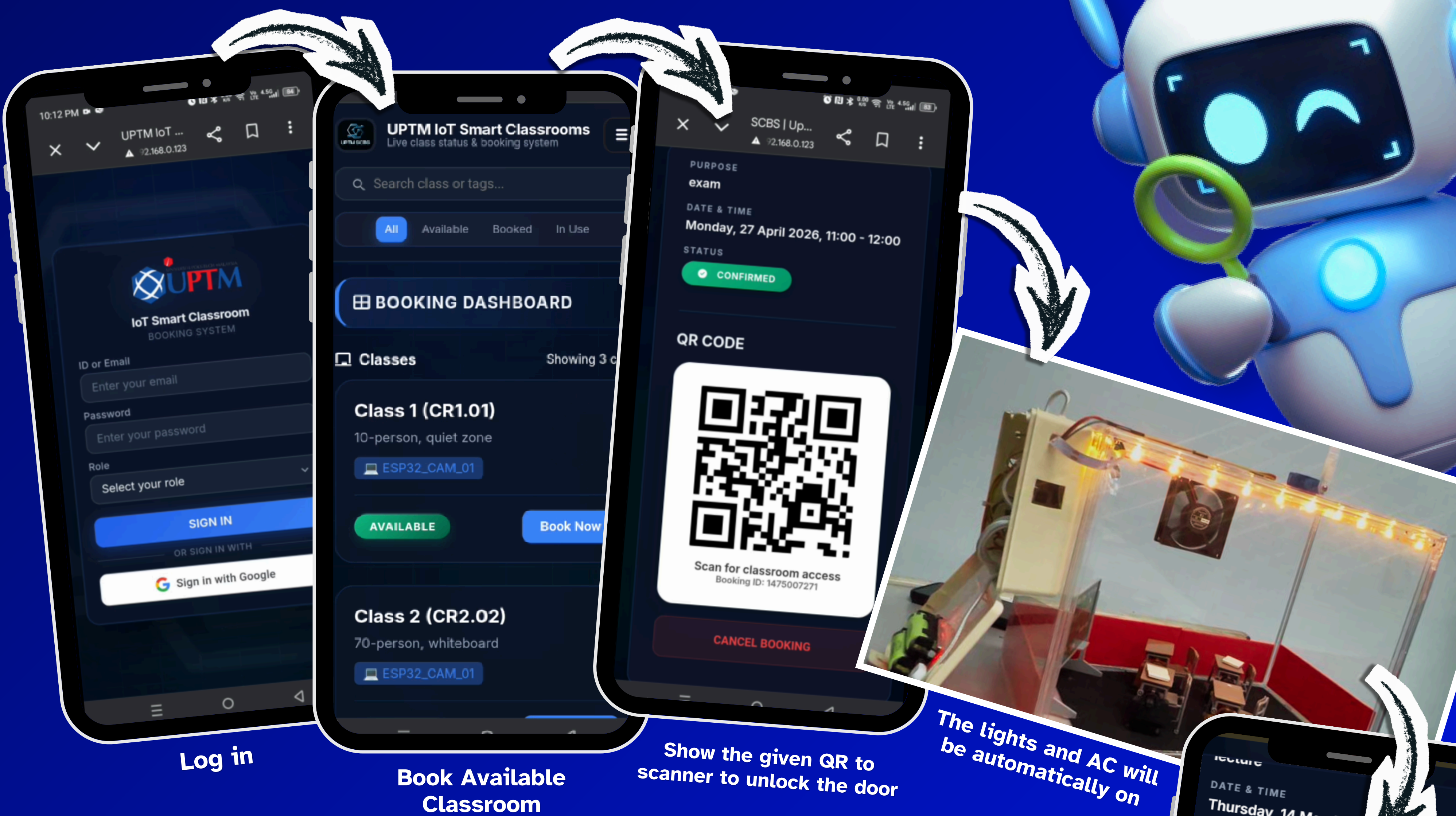
Saves Costs : Significantly reduces electricity bills via IoT energy automation.
- 

Improves Security : Restricts physical classroom access to authorized users only.
- 

Modernizes Campus : Establishes a scalable, tech-forward digital foundation for universi.
- 

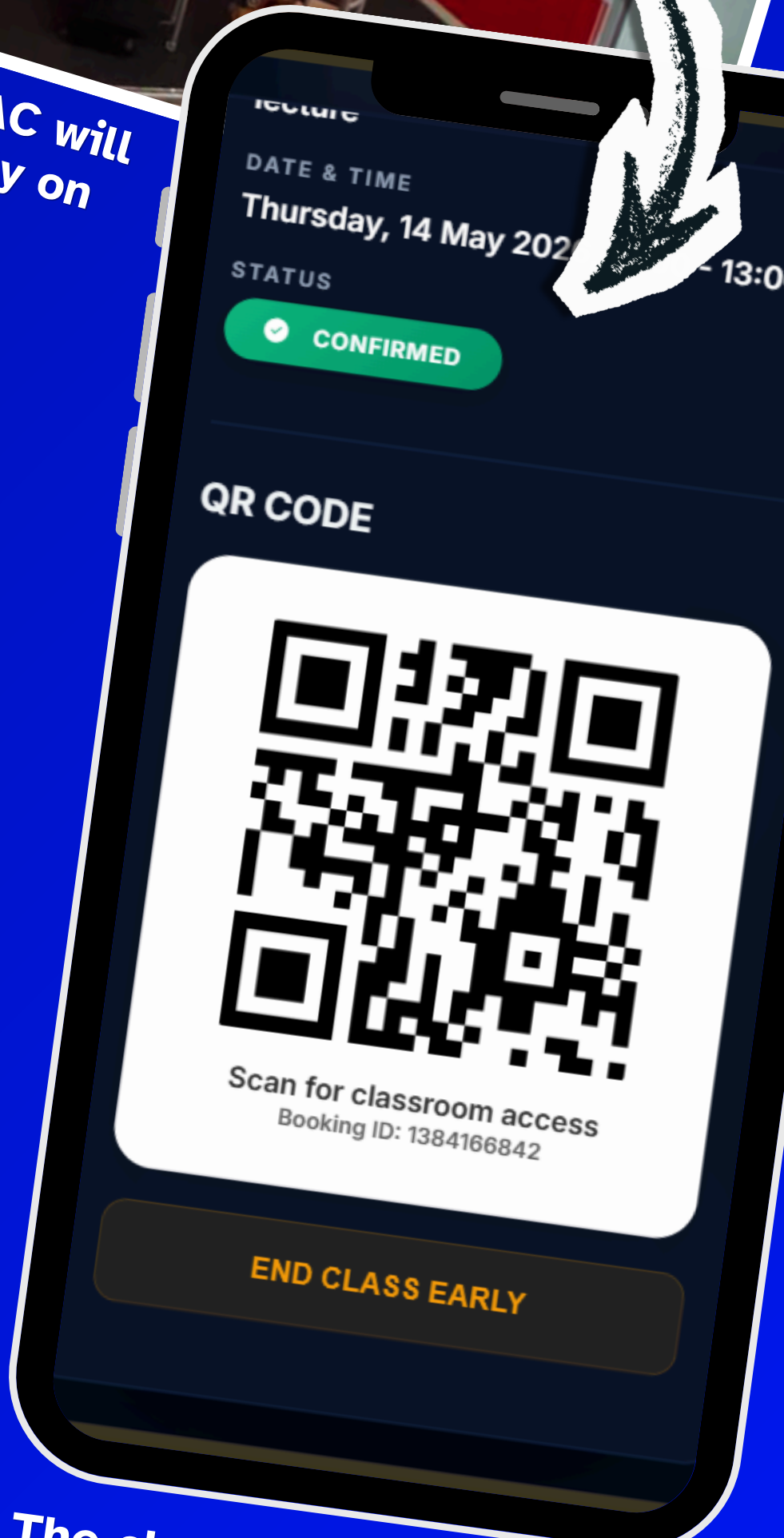
Drives Sustainability : Directly supports UN SDGs 7 (Clean Energy) and 12 (Responsible Consumption).

MODEL APPLICATIONS

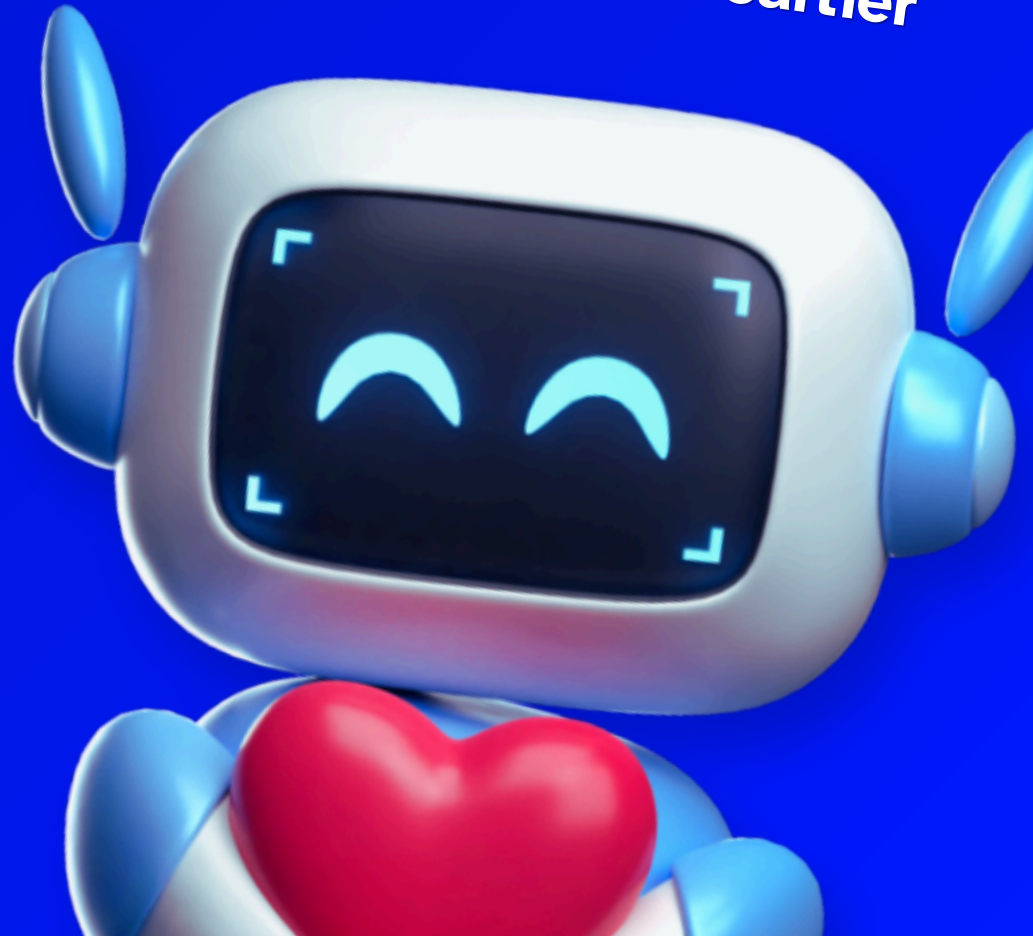


RESULTS & PERFORMANCE

FEATURE	PURPOSE	VALIDATION METHOD	RESULTS & PERFORMANCE
CENTRALIZED WEB BOOKING	To digitize scheduling and eliminate booking conflicts.	User Acceptance Test (UAT) with 50+ campus users.	100% prevention of double-bookings; 90% user preference over manual logs.
IOT ENERGY AUTOMATION	To reduce utility wastage in empty classrooms.	Live hardware testing with ESP32 & PIR sensors.	Reliable auto-shutdown of lights and AC triggered precisely after 15 minutes of inactivity.
QR CODE ACCESS	To secure entry and track physical attendance.	Scanner and door-lock (servo) actuation tests.	Fast, accurate scanning of encrypted tokens with zero unauthorized entries recorded.
ADMIN DASHBOARD	To provide oversight on room usage and occupancy.	System and database load testing by admins.	Seamless real-time updates of room statuses and accurate generation of usage reports.



The class can be end earlier



ACHIEVEMENTS



COPYRIGHT NO
LY2026W02824



PATENT NO
P12026002881